SYS

TIM2

stm32F746

global interrupt

Software Architecture

interrupt based process at 40us

Communication Profile

Command words
Status words
References

CAN-BUS

RX

TX

Global State Machine

Global control
Protections
Profile communication

Inverter Control State Machine

Inverter state machine
Control mode management
References Management
Protections
Errors Management

Speed/Torque - Vector Control

Torque/current control
Speed control
SV-PWM

SV-PWM

ADCs/HW protections/GPIOs

ADCs Acquisitions
MOSFET Driver Enabling

BEMF based model Observer

Rotor Phase estimation based
on BEMF model observer